Joint Physical Realization of Relational Extrema States: Sharp Resource Obstructions, Correlated Terminal Fibers, and Spectral Rigidity

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Abstract

A diffusion equation determines which temperature fields are physically attainable, but it does not by itself specify which observer-relative extrema relations and connected terminal resolutions can coexist in one field. We formulate this *joint realization* problem for selected extrema–frame foreground/background (EF–FG) relational predicates on finite thermal networks, retaining one common field and elder resolution while permitting separate intrinsic connected-region choices at distinct terminals. We first express shared actuation constraints through an exact support-function certificate. A rational weighted tree of maximum degree three exhibits a minimal three-query obstruction and a rigorously certified sharp total-power threshold of $108.32865113364069\ldots$; a two-switch full-power control realizes the threshold as equality and is unique almost everywhere. We then characterize exposed reachable-face dimensions through finite Krylov tie spaces and the dimensions of terminal cap fibers after affine equality constraints. On an eight-vertex tree, an optimal face is nonsemialgebraic, yet the pair of actual terminal caps has an exactly attainable correlated line-segment range. Changing a single conductance by one percent raises a terminal-difference Krylov rank from two to seven and collapses the corresponding optimal physical face to a point, while multiple valid terminal resolutions remain. The results give constructive realizations, global impossibility certificates, and a precise distinction between physical and relational nonuniqueness. They constitute mathematical physical realizations of EF–FG predicates, not hardware experiments or a claim of general superiority over classical control algorithms.

Keywords: extrema–frame semantics; graph heat equation; joint realizability; shared-control budget; dual certificate; Krylov rank; terminal resolution; nonsemialgebraic exposed face.

1 Introduction

A controlled temperature field can satisfy an individually specified target without satisfying several targets together. The difficulty becomes particularly visible when the targets are not point temperatures but *relational states*: an observer is dominated by a persistent local maximum, a designated terminal owns a separator, or a connected terminal region has a prescribed floor, cap, and membership flag. Each observable may be possible in isolation. Their juxtaposition is a valid physical claim only if all observables are evaluated on the *same* terminal temperature field.

We examine this issue in a finite graph realization of extrema–frame foreground/background (EF–FG) semantics. EF–FG specifies a superlevel-component elder rule, observer-relative saliency and ownership, spatial maximin gates, and intrinsic connected residual terminal regions. The foundational EF–FG I and II articles define the observer and spatial-frame outputs and their common-realization consistency requirements [1, 2]. The physical plant is conventional positive linear heat diffusion. Its novelty is *not* presumed: support-function duality, the bang–bang principle, finite-dimensional Krylov subspaces, and spectral graph perturbation are established ideas [3, 4, 6, 7]. What is investigated here is the exact coupling of these tools to a *joint,*

source-faithful relational realization problem, including the terminal caps and their correlations. Related continuous-time reachability and spectral decision problems have already been studied in substantially more general control-theoretic settings [5].

Contributions. The paper supplies four linked mathematical outcomes. First, it expresses physical common-witness feasibility in a dual form while keeping strict ownership inequalities distinct from weak terminal conditions. Second, on a nonsymmetric degree-three thermal tree, it computes a sharp resource threshold and its unique two-switch realizing control, together with witnesses for every proper subfamily of the three requested predicates. Third, it gives an active-tie Krylov formula for the dimension of an optimal physical face and, subject to a necessary relative-interior qualification, an exact cap-fiber dimension formula. Fourth, on a controlled pair of eight-vertex networks, it identifies a nonsemialgebraic optimum face with a *sharp correlated* cap segment and demonstrates how a one-percent conductance perturbation collapses that face to one physical field without removing the intrinsically multiple terminal-region choices.

The goal is therefore both a physical application and a mathematical test of the utility of relational semantics: the answers identify *what can be realized jointly, why a joint demand can fail, how much physical resource removes the failure, and which terminal outputs remain coupled.* All actuation examples are exact mathematical solutions of ideal network heat equations; no apparatus measurements are claimed.

The mathematical contribution and the model boundary

The word *realization* has two meanings in nearby literatures. In abstract EF–FG complexity theory, a combinatorial structure can be encoded by a small-radius scalar-field gadget. Here, by contrast, a *physical realization* is an actual solution of a constrained network heat equation, with nonnegative heater powers, one initial field, and one observation time. Neither type of theorem automatically implies the other. We use the EF–FG formalism as a prescribed output semantics, not as a substitute for the physical dynamics. The underlying definitions are developed in the foundational EF–FG I article [1] and extended in the EF–FG II article [2], especially for exact common-realization semantics, observer records, spatial frames, and correlated relational outputs.

More precisely, the full deterministic EF–FG state contains an observer component and a spatial-frame component, schematically $(PO_x(F), SF_x(F))$. We investigate *selected projections* of that state—bright ownership, existence of an intrinsic connected terminal resolution with a given membership flag, and the floor/cap data of chosen resolutions. We do not claim a universal physical realization theorem for every component of every categorical EF–FG output. This distinction is especially relevant when the joint query involves two terminals: the terminal regions are part of the common witness, but their identities need not be unique even after the physical field is fixed.

The paper is organized around three progressively finer questions. First, is a joint semantic demand possible under a given physical budget? Second, if it is possible, can we determine an exact least budget and a realizing control? Third, when many controls realize the optimum, what restrictions do they impose on the *joint* terminal tuples, rather than on their unrelated coordinate-wise ranges?

2 Physical model and source-faithful EF–FG queries

Let $G = (V, E)$ be finite, connected and undirected, with strictly positive edge conductances $c_{ij} = c_{ji}$. Its Laplacian is L, and $P_s = e^{-sL}$. For the permitted heater vertices $J \subseteq V$ and a pointwise shared power bound $Q > 0$, consider

$$\dot{T}(t) = -LT(t) + \sum_{j\in J} e_j u_j(t), \quad T(0) = Z > 0, \quad u_j(t) \geq 0, \quad \sum_{j\in J} u_j(t) \leq Q \text{ a.e.} \tag{1}$$

For observation time $\tau > 0$ and backward kernel time s, the final field is

$$F(u) = P_\tau Z + \sum_{j\in J} \int_0^\tau P_s e_j u_j(\tau - s)\, ds, \qquad \mathcal{K}_Q = \{F(u) : u \text{ obeys (1)}\}. \tag{2}$$

The set $\mathcal{K}_Q$ is compact and convex. Indeed, the control simplex is weak-star compact in the bounded L^∞ ball, the endpoint map is weak-star continuous, and the pointwise constraints are convex. Moreover P_s is a nonnegative stochastic matrix.

To state the required piece of EF–FG semantics self-containedly, define for vertices x, y

$$\kappa_F(x,y) = \max_{\pi: x \rightsquigarrow y} \min_{z \in \pi} F(z). \tag{3}$$

Local maxima of the superlevel filtration of F generate primitive birth components. On merging components, the elder rule retains the older (higher-birth) component, with a consistent resolution of equal-birth ties. A finite seed p has birth B_p and elder death $d(p)$. The spatial selected seeds have $B_p - d(p) > \Delta_{\text{seed}}$, and the rooted co-winner basin $\mathcal{B}_p$ is the connected component through p of the set on which p maximizes seed attachment. The two observer polarities are obtained using F and $-F$. Ownership is *bright* only if bright saliency exceeds its threshold and dark saliency does not; it is not a synonym for the inequality at an arbitrary vertex pair.

For a designated terminal $t \in \mathcal{B}_p$, put

$$s_t = \kappa_{F|\mathcal{B}_p}(p,t), \quad \alpha(v) = \kappa_{F|\mathcal{B}_p}(p,v), \quad \gamma_t(v) = \kappa_{F|\mathcal{B}_p}(t,v). \tag{4}$$

Vertices with $\alpha > s_t$ belong to the peak side and must be excluded from a terminal region; vertices with $\gamma_t > s_t$ belong to the terminal side; the remaining eligible vertices satisfy $\alpha = \gamma_t = s_t$. An *intrinsic residual terminal region* $R \in \mathfrak{R}^{\text{res}}_{p,t}(F)$ is connected, contains t and the forced terminal component, excludes the forced peak vertices, and selects any other vertices only from the tie zone. Its real-valued data are

$$\text{ResRel}_{p,t,x}(F,R) = \left(s_t,\ U(R) = \max_{z \in R} F(z),\ y = F(x),\ \chi_x = \mathbf{1}_{\{x \in R\}} \right). \tag{5}$$

We also retain $L(R) = \min_{z \in R} F(z)$, as the floor need not equal s_t for every admissible R. When the terminal classification margin is zero, the terminal-neutral band is determined by $L(R), U(R)$ and y; in particular, $y = L(R) = U(R)$ is neutral, whereas $y = L(R) < U(R)$ is dark.

Definition 2.1 (Joint source-native witness)**.** *For two terminals t_1, t_2 and a separator b, set $E_i(F)$ to mean $\exists R_i \in \mathfrak{R}^{\text{res}}_{p,t_i}(F)$ with $b \in R_i$. A* joint witness *for ownership O_Δ and E_1, E_2 consists of one admissible control u, its single field $F(u)$, one consistent elder choice, and an admissible region R_i for each terminal. No additional condition $R_1 \cap R_2 = \varnothing$ is imposed: it is not part of the intrinsic per-terminal definition.*

For the trees below we certify an invariant EF–FG semantic chamber in which p is the sole selected finite bright seed, $\mathcal{B}_p = V$, the essential elder component is fixed, and the two intrinsic queries have the *proved* simplifications

$$O_\Delta(F) \iff g^\top F > \Delta, \qquad E_i(F) \iff h_i^\top F \geq 0, \qquad g = e_p - e_w, \quad h_i = e_{t_i} - e_b. \tag{6}$$

The reductions in (6) are *conclusions of model-specific extrema and connectedness checks*; they must not be taken as the general definition of EF–FG.

2.1 What is retained from the intrinsic spatial semantics

The gate, floor, cap, query value, and membership flag carry different information. The gate s_t depends on the path geometry in the rooted basin, whereas $U(R)$ and $L(R)$ depend on the *selected connected region*. For terminal margin Δ_{term}, the source-native thresholds are

$$T^+_{\Delta_{\text{term}}}(L,U) = \max\{L + \Delta_{\text{term}},\ (L+U)/2\}, \qquad T^-_{\Delta_{\text{term}}}(L,U) = \min\{U - \Delta_{\text{term}},\ (L+U)/2\}. \tag{7}$$

A query y is dark below T^-, neutral in the closed band $[T^-, T^+]$, and bright above T^+. In particular, at zero terminal margin, if $y = L = s_t < U$, then $T^- = (L+U)/2 > L$, so the same terminal membership flag can be paired with a *dark* label; if $U = L = y$, its label is neutral. Both outcomes can arise from different admissible regions on one temperature field.

This distinction explains why the marginal statements "terminal i can be neutral" and "terminal j can have a large cap" cannot simply be multiplied: there must exist a *single* F, a consistent elder resolution, and regions R_i, R_j realizing the full tuples together. For example, imposing disjointness of R_1 and R_2 when both must contain the separator b would create an immediate contradiction; but that disjointness is an *additional global gluing contract*, not part of the intrinsic two-terminal relation analyzed here.

Remark 2.2 (Scope of the semantic chamber)**.** *The equivalent temperature inequalities in* (6) *are used only after verifying the selected birth atoms, essential component, co-winner basin, maximin gates, peak-side exclusions, terminal-side requirements, and connectedness of the displayed regions. At an unverified birth or gate tie, the simple affine conditions must not replace the original EF–FG construction. The semantic checks below enumerate the intrinsic connected candidates using the native gate, forced-side, and tie-zone rules rather than assuming the guard equivalence everywhere.*

3 Shared-control duality and exact joint infeasibility

For any $v \in \mathbb{R}^V$, write $\sigma_Q(v) = \max_{F \in \mathcal{K}_Q} v^\top F$. The control simplex yields the exact support formula

$$\boxed{\sigma_Q(v) = v^\top P_\tau Z + Q \int_0^\tau \max\left(0, \max_{j \in J} v^\top P_s e_j\right) ds.} \tag{8}$$

This is classical support optimization, but it is particularly useful when the h_i in (6) are *actual* terminal-membership guards rather than ad hoc temperature targets.

Proposition 3.1 (Common-witness dual certificate)**.** *Fix a chamber in which* (6) *holds. For $\lambda_i \geq 0$, let $v = g + \lambda_1 h_1 + \lambda_2 h_2$. If $\sigma_Q(v) \leq \Delta$, then the joint EF–FG query $O_\Delta \wedge E_1 \wedge E_2$ is impossible for every measurable control in* (1)*. Conversely, if one admissible F satisfies $g^\top F > \Delta$ and both $h_i^\top F \geq 0$, it is an exact common witness, irrespective of whether the R_i choices are unique.*

Proof. If E_i hold then $h_i^\top F \geq 0$, hence $g^\top F \leq v^\top F \leq \sigma_Q(v) \leq \Delta$, contradicting strict ownership. The converse follows from the verified equivalences (6) and the intrinsic connected-region witnesses. □

Remark 3.2. *For a general finite family of weak affine guards $AF \geq a$ and strict guards $CF > c$, the exact same-field feasibility test has two margins: $d_w = \max_{F \in \mathcal{K}_Q} \min_i((AF)_i - a_i)$, and, when $d_w \geq 0$, $d_s = \max_{F \in \mathcal{K}_Q, AF \geq a} \min_j((CF)_j - c_j)$. A witness exists precisely when $d_w \geq 0$ and $d_s > 0$. Minimax duality expresses these margins through* (8)*; zero-margin equality faces are not replaced by approximate tolerances. This is a convex-analytic characterization, not a claim of a terminating bit algorithm for arbitrary real spectral input.*

3.1 Exact support equality and complementary slackness

The support integral can be interpreted pointwise: in backward time s, the controller selects an action in $\mathcal{A} = \{0\} \cup \{e_j : j \in J\}$ with the largest value of $v^\top P_s a$; a nonzero winner is assigned the full available power. Tied winners permit convex sharing, but ties at isolated times do not alter the final field. Consequently the maximum in (8) is an *attained* global bound over all measurable controls, not a numerical discretization of time.

Lemma 3.3 (Equality in a two-terminal dual bound)**.** *Assume* (6)*, and let $\lambda_1, \lambda_2 > 0$. If a feasible field F satisfies $g^\top F = \sigma_Q(g + \lambda_1 h_1 + \lambda_2 h_2)$ and $h_i^\top F \geq 0$, then both $h_i^\top F = 0$ and the corresponding control attains the pointwise support maximizing action almost everywhere. If this action is unique except at finitely many switching instants, the realizing control is unique almost everywhere and the exposed face is a singleton.*

Proof. The nonnegative sum $\lambda_1 h_1^\top F + \lambda_2 h_2^\top F$ must vanish since $g^\top F \le (g + \lambda_1 h_1 + \lambda_2 h_2)^\top F \le \sigma_Q$. Positive multipliers force the two individual equalities. The difference between σ_Q and the actual dual value is the integral of a nonnegative pointwise control-allocation loss. An integral of a nonnegative measurable function is zero only when the loss vanishes almost everywhere. Uniqueness of the pointwise maximizing action therefore transfers to the control and its endpoint. □

A finite-dimensional certificate workflow. For a proposed active-port sequence, one solves the exact integral constraints for its switching times, actuation scale, and positive dual multipliers. The candidate is *not* accepted merely because a local solver converges. One must also prove (i) all required EF–FG guard identities on the same field; (ii) positivity of all dual multipliers; (iii) dominance of the proposed heater over *every* alternate heater and idle on the entire relevant time intervals; and (iv) a certified root enclosure. Steps (ii)–(iii) convert the local candidate into a global proof against arbitrary measurable controls. We implement this workflow in the three networks below.

4 A sharp minimal resource obstruction on a nonsymmetric tree

Consider the seven-vertex tree in Figure 1, with vertex order $(r, w, p, b, t_1, c, t_2)$ and rational conductances

$$(c_{rw}, c_{wp}, c_{pb}, c_{bt_1}, c_{bc}, c_{ct_2}) = (1.01, 1.02, 0.97, 1.03, 0.96, 1.04). \tag{9}$$

Take $Z = (25, 5, 15, 11, 10.8, 11.5, 10.8)$, $\tau = 0.01$, heaters p, t_1, t_2, observer parameter $\theta = 0.4$, spatial seed threshold 5, terminal margin 0, and strict ownership margin $\Delta = 10.16$.

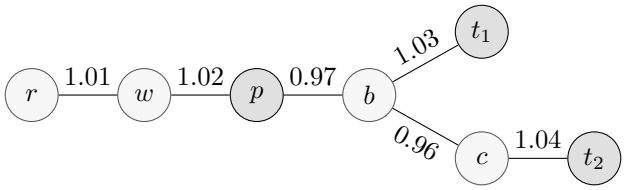


Figure 1: Seven-vertex nonsymmetric rational thermal tree. Shaded nodes are controllable heater ports. Both terminal queries concern the same separator b and final field.

A uniform positive-semigroup/Taylor interval argument verifies, for all admissible $Q \le 109$, that r remains essential and p the only selected finite bright seed with $\mathcal{B}_p = V$. The ownership and terminal-region conditions are exactly (6); for terminal t_2 the intervening node c is forced into every R_2 containing b. See Appendix A for the model-specific certification.

Define $K(\alpha, \beta) = \int_\alpha^\beta P_s \, ds$ and, for $0 < a < b < \tau$,

$$F(Q, a, b) = P_\tau Z + Q\left[K(0, a)e_p + K(a, b)e_{t_1} + K(b, \tau)e_{t_2}\right]. \tag{10}$$

Here the source order is p, t_1, t_2 in *backward* kernel time, equivalently t_2, t_1, p in real physical time.

Theorem 4.1 (Sharp resource threshold and unique realizing schedule)**.** *There exist exact real numbers Q_*, a_*, b_* with*

$$\begin{aligned} 108.32865113354069 < Q_* &< 108.32865113374070, \\ a_* &= 0.00555636232738007475\ldots, \\ b_* &= 0.00779845184741640996\ldots, \end{aligned}$$

uniquely specified in the certified local root box by

$$g^\top F(Q_*, a_*, b_*) = 10.16, \qquad h_1^\top F(Q_*, a_*, b_*) = h_2^\top F(Q_*, a_*, b_*) = 0. \tag{11}$$

For the actual source-native EF–FG queries of this graph and every *available pointwise budget* $Q \geq 0$*,*

$$\boxed{O_{10.16} \wedge E_1 \wedge E_2 \text{ is physically realizable} \iff Q > Q_*.} \tag{12}$$

At the critical budget, the solution of the associated weak *Bright problem is the two-switch control in* (10)*, unique up to a set of times of measure zero; the strict Bright condition fails by equality.*

Proof. The three equations in (11) are analytic in (Q, a, b). A rational Taylor expansion of P_s through degree 42, a rational approximate inverse Jacobian, and the contraction bounds listed in Appendix A give existence and uniqueness of the exact root within an explicit rational cube. Set $v = g + \lambda_1 h_1 + \lambda_2 h_2$, where λ_1, λ_2 make the source gains equal at a_* (ports p, t_1) and at b_* (ports t_1, t_2). Exact interval bounds give $0.989 < \lambda_1 < 0.991$ and $0.973 < \lambda_2 < 0.976$.

Let $D_1(s) = v^\top P_s(e_p - e_{t_1})$ and $D_2(s) = v^\top P_s(e_{t_1} - e_{t_2})$. The full-interval certified derivative estimates give $D_1'(s) < 0$ and $D_2'(s) < 0$, with $D_1(a_*) = D_2(b_*) = 0$ and $a_* < b_*$. It follows that the maximizing heater is p before a_*, t_1 between a_* and b_*, and t_2 afterward; both heater gains remain strictly positive, excluding the idle action. Thus (10) at Q_* attains the *global* support $\sigma_{Q_*}(v)$. The two terminal contrasts in (11) are exactly zero, so

$$\max_{F \in \mathcal{K}_{Q_*},\ h_i^\top F \geq 0} g^\top F = 10.16.$$

Proposition 3.1 proves impossibility for $Q \leq Q_*$ (reachable sets are nested) and shows that equality is not sufficient for strict ownership. Since the optimal action is strictly unique except at the two switching instants, any control attaining the support agrees with the stated two-switch schedule almost everywhere.

For any $Q > Q_*$, choose $Q' \in (Q_*, \min\{Q, 109\})$. Apply the same switching times at power Q'. By linearity, from $h_i^\top F(Q_*) = 0$ and $h_i^\top P_\tau Z < 0$ we get $h_i^\top F(Q') > 0$; similarly $g^\top F(Q') > 10.16$. The bound $Q' < 109$ preserves the verified EF–FG chamber. This produces a genuine connected-region common witness for every larger available budget. □

At $Q = 109$, the same schedule gives $g^\top F \approx 10.163699289185$, $h_1^\top F \approx 0.001474729213$, and $h_2^\top F \approx 0.001444220901$, all on the required strict sides. Thus the theorem distinguishes an approximate numerical target from an exact resource threshold.

Proposition 4.2 (Minimal three-query obstruction)**.** *At* $Q = 100$*, the family* $\{O_{10.16}, E_1, E_2\}$ *is minimally unrealizable: every proper subfamily has a common physical witness, whereas the three-query family has none.*

Proof. Impossibility is Theorem 4.1. Every proper pair is witnessed by a constant rational power allocation $u = (u_p, u_{t_1}, u_{t_2})$: $O \wedge E_1$ by $(70, 30, 0)$, $O \wedge E_2$ by $(70, 0, 30)$, and $E_1 \wedge E_2$ by $(0, 50, 50)$. Single predicates are inherited from these pair witnesses. The exact endpoint inequalities follow from rational-Taylor interval evaluation of (2); their certified margins are summarized in Table 1. No three independent final fields are recombined. □

Family	u_p	u_{t_1}	u_{t_2}	$g^\top F$	$h_1^\top F$	$h_2^\top F$
O	100	0	0	10.54820	−0.24272	−0.23781
E_1, E_2	0	50	50	9.56309	0.25693	0.26183
O, E_1	70	30	0	10.25267	0.05565	−0.23790
O, E_2	70	0	30	10.25266	−0.24129	0.06207

Table 1: Mathematical physical witnesses for every proper subfamily at $Q = 100$. Values shown are rounded, while signs are separately certified.

Even at the unique critical physical field there are different admissible terminal resolutions. Specifically, the regions containing b for the first terminal are $\{b, t_1\}$, $\{b, c, t_1\}$, and $\{b, c, t_1, t_2\}$; for the second terminal they are $\{b, c, t_2\}$ and $\{b, c, t_1, t_2\}$. There are six valid per-terminal pairs evaluated on the *same* final field and elder resolution, with two neutral–dark and four dark–dark label combinations. Physical control uniqueness therefore does not force semantic uniqueness.

4.1 From controls to one physical state: the seven-vertex case

The resource threshold separates three logically distinct statements: there are individual physical witnesses, there are witnesses for every *proper* subfamily, and there is no witness for the full family at $Q = 100$. The first two follow from the explicitly displayed constant controls; the third requires a bound valid for *every* possible time-varying allocation. Theorem 4.1 proves the third without restricting the control to a chosen number of switches.

In forward physical time the critical controller activates t_2 on $[0, \tau - b_*)$, then t_1 on $[\tau - b_*, \tau - a_*)$, and then p on $[\tau - a_*, \tau]$, with power Q_* at each active port. The approximate switches are 0.00220154815258 and 0.00444363767262. The exact values are defined by (11); these decimals should not be used as a substitute for the exact construction when verifying terminal equalities. Both switching instants and all three heater allocations are physically admissible in the ideal graph model. No instantaneous negative heating, off-budget pulse, or different temperature field for the two terminals is used.

The two terminals probe the *same* separator b but different routes. For t_1, the connected terminal region can include b without passing through c. For t_2, the path $t_2 - c - b$ makes c mandatory whenever b is included. In the critical field $F(c) > F(b)$. Thus the source-native $\chi_b = 1$ region for t_2 has a cap strictly above its floor, whereas t_1 admits the neutral choice $\{b, t_1\}$. The six possible pairs of b-containing resolutions split into two neutral–dark and four dark–dark cases. This difference is a direct geometric statement about connected regions, not an artifact of the heating optimization.

Remark 4.3 (What the power number does and does not mean)**.** *The value Q_* is a sharp* pointwise aggregate heater-power *threshold for the specified ideal graph, initial data, observation time, and EF–FG predicates. It is not a universal material energy scale, not an experimentally measured wattage, and not a bound on the time integral of power. Converting it to physical watts requires a choice of thermal capacitances, conductance units, and a dimensional normalization of* (1)*.*

5 A finite spectral test for the dimension of an optimal physical fiber

Let $\mathcal{A} = \{0\} \cup \{e_j : j \in J\}$ be the allowed instantaneous actions (the zero vector denotes idle), and let Π_λ be the orthogonal spectral projectors of L. For a dual direction v set $k_j(s) = v^\top P_s a_j$. A pair (i, j) is called an *active tie* if both gains are equal to the *maximum* over all allowed actions on a nonempty open time interval. The qualification "maximum" is essential: a tie between two suboptimal ports cannot create an optimal-face direction.

Theorem 5.1 (Active-tie Krylov criterion)**.** *Suppose L has d distinct eigenvalues. The following conditions are equivalent for any two actions a_i, a_j:*

(i) $v^\top P_s(a_i - a_j) = 0$ *on a set of positive measure;*

(ii) this gain difference vanishes for every real s;

(iii) $v^\top L^k(a_i - a_j) = 0$ *for $k = 0, \dots, d-1$;*

(iv) v is orthogonal to every $\Pi_\lambda(a_i - a_j)$.

For the face-dimension statement below, assume explicitly that $Q > 0$. Let $\mathcal{T}_v$ collect the active tied pairs*, and define*

$$S_v = \operatorname{span}\{\Pi_\lambda(a_i - a_j) : (i, j) \in \mathcal{T}_v,\ \lambda \in \operatorname{spec}(L)\}. \tag{13}$$

For the exposed face $\mathcal{K}_{Q,v} = \{F \in \mathcal{K}_Q : v^\top F = \sigma_Q(v)\}$,

$$\operatorname{aff}(\mathcal{K}_{Q,v}) = F_0 + S_v, \qquad \dim \mathcal{K}_{Q,v} = \dim S_v, \tag{14}$$

for any $F_0 \in \mathcal{K}_{Q,v}$. In particular, the face is a singleton precisely when there is no nontrivial active tie.

Proof. The gain difference is the real-analytic exponential sum $\sum_\lambda e^{-\lambda s} v^\top \Pi_\lambda (a_i - a_j)$. An analytic function vanishing on a positive-measure set is identically zero. Linear independence of exponentials and the Vandermonde system from its derivatives at zero give equivalence of the four conditions.

For the face formula, the assumption $Q > 0$ allows nonzero bidirectional transfers of power between pointwise tied actions. Equality in the support integral (8) forces any optimal control to put its instantaneous mass only on actions tied for the pointwise maximum (including idle if it ties). Differences between any two such controls therefore lie in S_v. Conversely, on an open active-tie interval choose a baseline that assigns strictly positive mass to each tied action, and make small positive and negative measurable perturbations that transfer mass between those actions. The linear span of all resulting endpoint differences contains $\operatorname{span}\{P_s(a_i - a_j) : s \in I\}$. Analyticity and linear independence of $e^{-\lambda s}$ imply this span equals $\operatorname{span}\{\Pi_\lambda(a_i - a_j)\}_\lambda$. Summing over the finitely many active-tie classes gives S_v, proving both inclusions. Nonidentical analytic gains can coincide only at isolated instants on $[0, \tau]$ and hence contribute no face dimension. □

Corollary 5.2 (Rank of the cap image after terminal equalities)**.** *Assume $Q > 0$. Let $HF = \eta$ encode terminal equalities and suppose* $\operatorname{relint}(\mathcal{K}_{Q,v}) \cap \{F : HF = \eta\} \neq \varnothing$. *Then*

$$\operatorname{aff}(\mathcal{K}_{Q,v} \cap \{HF = \eta\}) = F_0 + (S_v \cap \ker H). \tag{15}$$

Let $\mathcal{U}$ be a nonempty relatively open subset of this sliced face on which a prescribed connected-region selection stays admissible and the maximizer of every cap is a fixed vertex. On $\mathcal{U}$ the cap vector is a linear map CF, and its attainable image has affine dimension $\operatorname{rank}(C|_{S_v \cap \ker H})$.

Proof. The relative-interior intersection contains a relative neighborhood of F_0 within the affine slice $F_0 + S_v \cap \ker H$, so that slice is precisely the affine hull. Since $\mathcal{U}$ is relatively open and nonempty in the affine slice, $\operatorname{aff}(\mathcal{U})$ is that whole slice; a linear image has dimension equal to the rank of its restricted direction space. □

Remark 5.3. *The positivity hypothesis $Q > 0$ is necessary for the face-dimension formula: at $Q = 0$ the reachable set is a singleton even if pointwise actions tie and S_v is nonzero. Without the relative-interior hypothesis, the rank can be only an upper bound; a boundary slice may be smaller or empty. A fixed region R alone does not fix $U(R)$ as a linear function when its maximizing vertex changes; the active-cap chamber qualification is essential. Caps of* different *connected-region branches must never be multiplied as independent coordinate ranges.*

5.1 Why active ties, not graph automorphisms alone, govern optimal fibers

Theorem 5.1 distinguishes a symmetry of the physical network from a symmetry that is actually *visible in the dual objective.* A graph automorphism may exchange two sources, but if the dual direction is not invariant under the exchange, their gains need not tie. Conversely, equal dual gains can be certified by vanishing Krylov moments even in the absence of an obvious vertex automorphism.

For a candidate pair of heater actions, form the $n \times n$ Krylov matrix $[d, Ld, \ldots, L^{n-1}d]$, where $d = a_i - a_j$. Its column rank determines how large the subspace of endpoint perturbations *could* be under sustained source sharing, whereas the test $v^\top L^k d = 0$ determines whether a specified dual direction actually observes them as tied. Only pairs that also attain the *maximum* gain on a nonempty open interval contribute to the exposed face $\mathcal{K}_{Q,v}$. If terminal equality constraints are then imposed, the permissible perturbations lie in $S_v \cap \ker H$.

The cap image needs one further check: the selected connected regions must remain admissible on an open portion of that affine slice, and their cap-maximizing vertices must not change. With these conditions the matrix rank in Corollary 5.2 is exact; without them a number obtained by rank alone can overstate the actual cap freedom. This qualification is central to interpreting EF–FG data as *joint relations* rather than independent coordinate intervals.

6 A nonsemialgebraic optimal face with a sharp correlated cap segment

Consider the symmetric eight-vertex tree in Figure 2. Its vertex order is $(r, w, p, b, x_1, t_1, x_2, t_2)$; the backbone conductances $(r, w), (w, p), (p, b)$ are $(1.01, 1.02, 0.97)$, and all four arm conductances equal 1. Let

$$Z = (25, 5, 15, 11, 11.5, 10.8, 11.5, 10.8), \quad \tau = 0.01, \quad Q = 100, \tag{16}$$

with heater ports p, t_1, t_2. The EF–FG parameters are $\theta = 0.4$, spatial seed threshold 5 and terminal margin 0. Exact uniform bounds again certify (6) with both arms satisfying $F(x_i) > F(b)$ for all permitted controls.

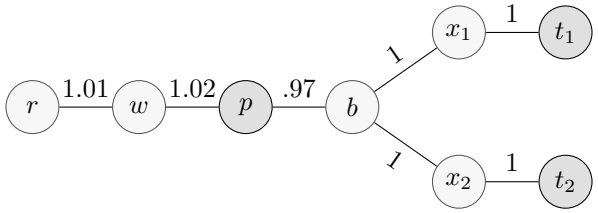


Figure 2: An eight-node thermal tree with exchangeable terminal arms. Replacing only the x_2t_2 conductance 1 by 1.01 yields the nonsymmetric comparison in Section 7.

The arm-swap symmetry preserves both the objective $g^\top F = F_p - F_w$ and the control simplex and exchanges the two terminal inequalities. Averaging a feasible control with its swapped counterpart preserves $g^\top F$ and turns $h_1^\top F, h_2^\top F$ into their common average. This exact group-averaging argument reduces the maximum of g subject to both real terminal guards to a symmetric two-port problem with $e_T = (e_{t_1} + e_{t_2})/2$ and $h = (h_1 + h_2)/2$. It does *not* average the connected regions themselves.

Lemma 6.1 (Finite-group reduction for affine source-native guards)**.** *Let a finite group Γ act by vertex permutations preserving L, Z, the permitted control set, and the aggregate power bound. Suppose the objective $g^\top F$ is invariant and the affine guards $h_i^\top F \geq 0$ form a permutation orbit under Γ. Then an optimizer of $g^\top F$ subject to all guards can be chosen Γ-invariant. On this invariant subspace all orbit guards have the same value, so the family reduces to their average.*

Proof. Given an admissible control u, average its orbit of permuted controls: $\bar{u} = |\Gamma|^{-1} \sum_{\gamma \in \Gamma} \gamma u$. Nonnegativity and the pointwise aggregate power budget are preserved by convexity. The endpoint map is equivariant because P_s commutes with Γ and Z is invariant, hence $F(\bar{u}) = |\Gamma|^{-1} \sum_\gamma \gamma F(u)$. The invariant objective is unchanged. Since every orbit of a satisfied guard stays satisfied under the corresponding permutation, the average obeys each guard. Finally each guard has the same value on a symmetric field, so it is equivalent there to the orbit average. □

The lemma concerns *affine temperature guards after a verified semantic reduction.* The connected sets R_i themselves are not averaged into a fictitious connected set: once the common symmetric field is obtained, each genuine V18 terminal region is selected and checked on that field. In the eight-node tree, the group has two elements, exchanging the two terminal arms and leaving the observer p fixed.

Theorem 6.2 (Sharp two-terminal obstruction and nonsemialgebraic optimal face)**.** *For the symmetric model* (16)*, the exact maximum of $g^\top F$ subject to E_1 and E_2 is*

$$G_* = 10.072177752864946627341898\ldots, \qquad 10.07217775286 < G_* < 10.07217775287. \tag{17}$$

In particular $O_{10.07} \wedge E_1 \wedge E_2$ is realizable, while $O_{10.08} \wedge E_1 \wedge E_2$ is not; every proper subfamily of the latter three-query family is realizable at $Q = 100$. The exposed physical support face *of the corresponding optimal dual direction is not a semialgebraic set.*

Proof. Define the symmetric one-switch field $F_s = P_\tau Z + 100[K(0,s)e_p + K(s,\tau)e_T]$. The scalar $H(s) = h^\top F_s$ decreases strictly from positive to negative and has one exact root

$$s_* = 0.0051299779322837138844\ldots.$$

Choose the positive multiplier $\lambda_* = -g^\top P_{s_*}(e_p - e_T)/[h^\top P_{s_*}(e_p - e_T)] \approx 1.96017386923015$. Certified whole-interval bounds for $v = g + \lambda_* h$ yield a strictly decreasing source-gain difference, unique zero s_*, and positive gains for both ports. Proposition 3.1 then gives the global bound $g^\top F \le g^\top F_{s_*} = G_*$ whenever $h_1^\top F, h_2^\top F \ge 0$. The symmetric source allocation $u_{t_1} = u_{t_2} = 50$ in backward time $s \in [s_*, \tau]$, followed by $p = 100$ on $[0, s_*]$, realizes both terminal equalities and this value. The proper subfamily witnesses are the constant source triples $(100, 0, 0)$, $(0, 50, 50)$, $(70, 30, 0)$, and $(70, 0, 30)$.

To prove the semialgebraic claim, note that on the support face the source p uniquely wins for $s < s_*$, while t_1, t_2 are tied as maximum for $s > s_*$. Thus the total terminal heating there is 100, but its signed allocation $q(s) = u_{t_1}(\tau - s) - u_{t_2}(\tau - s)$ can vary through all measurable values $[-100, 100]$ on $[s_*, \tau]$. The antisymmetric arm Laplacian is $B = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, with eigenvalues $a = (3 - \sqrt{5})/2$ and $b = (3 + \sqrt{5})/2$, whose ratio $(7 + 3\sqrt{5})/2$ is irrational. The linear projection onto the two antisymmetric spectral modes is, up to a fixed translation, exactly

$$\left\{ \left(\int_{s_*}^{\tau} e^{-as} q(s) ds,\ \int_{s_*}^{\tau} e^{-bs} q(s) ds \right) : |q(s)| \le 100 \right\}. \tag{18}$$

The boundary of this two-moment body contains an analytic irrational-power arc $Y = \alpha + \beta(\gamma + \delta X)^{b/a}$ on an open interval (after converting q to a nonnegative bounded input). Such an arc cannot satisfy any nonzero polynomial relation: the functions $x^{i+j(b/a)}$ have pairwise distinct exponents, and are linearly independent on a positive interval. An analytic semialgebraic arc would satisfy a polynomial relation [9]. Hence the projection (18) is not semialgebraic. If the exposed face were semialgebraic, its linear image would be semialgebraic by the Tarski–Seidenberg theorem, a contradiction. □

The nonsemialgebraic face of Theorem 6.2 contains optimal states that do *not* satisfy both terminal constraints. The genuine *jointly feasible* optimum fiber is obtained by intersecting it with the two equalities $h_1^\top F = h_2^\top F = 0$. This is the distinction required when reporting physically correlated relational outputs.

6.1 Why the optimal physical face can be nonsemialgebraic

For clarity, the source of the irrational-power boundary in (18) can be made explicit, rather than left as a general algebraic-geometric assertion. Fix $A < B$, rates $0 < a < b$, and a scalar input $0 \le u(s) \le Q$ on $[A, B]$, and put

$$X = \int_A^B e^{-as} u(s) ds, \qquad Y = \int_A^B e^{-bs} u(s) ds.$$

For a prescribed X, the largest Y is obtained by allocating full power at the *earliest* values of s until the required X is reached. Indeed, the ratio of the Y-kernel to the X-kernel, $e^{-(b-a)s}$, strictly decreases with s; shifting an infinitesimal amount of weighted input from a later to an earlier time increases Y without changing X. The rearrangement is exact over measurable inputs, including arbitrary fractional controls.

Thus the upper boundary is achieved by $u = Q$ on $[A, A + \ell]$ and zero afterward. Eliminating ℓ yields

$$Y_{\max}(X) = \frac{Q}{b} \left[e^{-bA} - \left(e^{-aA} - \frac{aX}{Q} \right)^{b/a} \right], \quad 0 < X < \frac{Q}{a}(e^{-aA} - e^{-aB}). \tag{19}$$

If b/a is irrational, this real-analytic arc is not the zero set of any nonzero polynomial in (X, Y): a polynomial relation would produce a finite linear combination of distinct real powers $z^{i+j(b/a)}$,

which are linearly independent on any positive interval. Since an analytic semialgebraic arc is Nash, this boundary and hence the associated two-dimensional moment body are not semialgebraic. Conversely a rational exponent gives an algebraic power relation and a semialgebraic moment body. This concrete boundary is sufficient for the present proof; it is not a statement that every irrational-spectrum reachability query is undecidable.

The antisymmetric source allocation of our physical tree satisfies $-Q \le q \le Q$. Replacing q by $2u - Q$ is an affine translation and scaling of its two modal moments. Therefore the preceding argument applies unchanged to the exact linear image of the physical optimal support face. The use of a *linear* image matters: if the whole face were semialgebraic, this image would be semialgebraic by projection, contradicting (19).

Theorem 6.3 (Exactly attainable correlated terminal caps)**.** *Fix on the optimal jointly feasible fiber the two intrinsic admissible connected regions*

$$R_1 = \{b, x_1, t_1\}, \qquad R_2 = \{b, x_2, t_2\}, \tag{20}$$

whose caps are $U_1 = F(x_1)$ and $U_2 = F(x_2)$. Then the complete attainable cap set is

$$\boxed{(U_1, U_2) = (C + z/2, C - z/2), \quad -\delta \le z \le \delta,} \tag{21}$$

where $C = 11.490213858830921509\ldots$ and $\delta = 0.000584009198972739483\ldots$. The endpoints are realized by full-power two-switch schedules, and every intermediate pair by admissible convex mixing. In particular, the two individual cap intervals may not *be combined independently.*

Proof. Every optimizer obeying both terminal guards has $h_1 = h_2 = 0$ because the two dual multipliers are strictly positive and g attains the support bound. The symmetric component of the final field is uniquely fixed by the winning action p for $s < s_*$ and total terminal source 100 for $s > s_*$. Therefore the cap sum is $2C$.

Let q be the signed difference of terminal heating on $[s_*, \tau]$. For the two antisymmetric arm eigenvalues $a < b$ in the preceding proof, set

$$\begin{aligned} k_y(s) &= \frac{(b-1)e^{-as} + (1-a)e^{-bs}}{b-a} > 0, \\ k_x(s) &= \frac{e^{-as} - e^{-bs}}{b-a} > 0. \end{aligned} \tag{22}$$

The terminal equality is $\int_{s_*}^{\tau} k_y(s)q(s)ds = 0$ and the cap difference is $U_1 - U_2 = \int_{s_*}^{\tau} k_x(s)q(s)ds$. The ratio k_x/k_y strictly increases with s: setting $z = e^{-(b-a)s}$ turns it into $(1-z)/[(b-1)+(1-a)z]$.

Let $c \in (s_*, \tau)$ be the unique weighted median satisfying

$$\int_{s_*}^{c} k_y(s)ds = \int_{c}^{\tau} k_y(s)ds.$$

Among all $|q| \le 100$ with zero k_y-moment, the rearrangement inequality (or the pointwise sign of $k_x - \mu k_y$) gives the sharp extremes $q = \pm 100$ on $[s_*, c]$ and the opposite sign on $[c, \tau]$. Thus

$$\delta = 100\left(\int_{c}^{\tau} k_x(s)ds - \int_{s_*}^{c} k_x(s)ds\right) > 0, \quad c = 0.00756204666625562686\ldots.$$

These controls satisfy the terminal equality exactly and yield the two endpoints of (21). Convex interpolation gives every intermediate z without changing the optimal objective or terminal equalities. Since $F(x_i) > F(b) = F(t_i)$ in the certified EF–FG chamber, (20) are valid intrinsic connected terminal regions, their floors are $F(b)$, and their caps are $F(x_i)$, as claimed. □

For the symmetric optimum one obtains $U_1 = U_2 = C$. At a two-switch endpoint the actual field has $U_1 \approx 11.4899218542$ and $U_2 \approx 11.4905058634$; both have the same $g^\top F = G_*$ and terminal equalities. The result is a physical explanation of why a pair of individually possible terminal cap values need not be jointly possible.

6.2 An exact relational obstruction inside an optimal face

The line segment in (21) gives more information than an inequality bounding each terminal separately. For example, the marginal values $U_1 = C + \delta/2$ and $U_2 = C + \delta/2$ are both physically realizable *individually* at the optimal Bright score and with both terminal membership guards. Nevertheless the pair $(C + \delta/2, C + \delta/2)$ is impossible: it violates the exact joint invariant $U_1 + U_2 = 2C$.

Similarly, the endpoint $(C + \delta/2, C - \delta/2)$ is witnessed by one full-power terminal switching order, and the reversed endpoint by interchanging the two arms. Their midpoint has a symmetric power-sharing realization. Every intermediate pair is obtained by convexly mixing the two endpoint controls, which is admissible because the power constraint is pointwise convex. This constructive argument proves both *necessity* and *sufficiency* of the correlation law, with no relaxation to a rectangular product of marginal cap intervals.

Corollary 6.4 (A minimal correlated-cap obstruction)**.** *On the jointly feasible optimal fiber of Theorem 6.3, let A assert $U_1 = C + \delta/2$ and B assert $U_2 = C + \delta/2$, for the fixed true regions* (20)*. Each assertion separately has a physical joint-terminal witness attaining G_*, but $A \wedge B$ has none on the same final field.*

Proof. The two endpoints of (21) witness each individual assertion. Their simultaneous conjunction would require $U_1 + U_2 = 2C + \delta > 2C$, contradicting the sharp cap invariant. □

This is a second kind of obstruction. Theorem 4.1 concerns the *resources required to obtain joint predicates*; Corollary 6.4 concerns the *coupling of two numerical relational outputs after physical optimality and terminal membership have already been achieved.* Neither is obtained by performing independent optimizations of the two terminal caps.

7 Conductance perturbation and relational rigidity

Keep the data of Section 6, changing only the conductance $c_{x_2t_2} = 1$ to $101/100$. The tree topology, allowable heaters, initial field, and two intrinsic terminal requests are unchanged. Put $d = e_{t_1} - e_{t_2}$ and $\mathcal{C}_d = [d, Ld, \dots, L^7 d]$.

Theorem 7.1 (Exact Krylov rank jump and collapse of the optimum face)**.** *The symmetric graph has* $\operatorname{rank} \mathcal{C}_d = 2$*, while the one-percent-perturbed graph has* $\operatorname{rank} \mathcal{C}_d = 7$*. For the latter graph a 7×7 subdeterminant, using rows $2, \dots, 8$ (one-based vertex indexing) and columns $d, \dots, L^6 d$, equals*

$$\frac{12710725656263913231}{125000000000000000000000000000} \neq 0.$$

For the perturbed network at $Q = 100$, the sharp joint-terminal Bright maximum is

$$G_{\mathrm{pert}} = 10.0722304026414387016940\ldots. \tag{23}$$

Its global dual-optimal physical field is unique; a pure full-power two-switch schedule, in forward physical time $t_1 \to t_2 \to p$, realizes it with both terminal equality conditions. The pair of caps corresponding to (20) *is uniquely*

$$(U_1, U_2) = (11.4905058628528755\ldots,\ 11.4898687382819038\ldots). \tag{24}$$

Nevertheless, each terminal still admits multiple distinct connected R_i choices, so the source-native complete ResRel *relation is not a singleton.*

Proof. The stated ranks and minor follow by exact rational multiplication of the two graph Laplacians; the vector d has coordinate sum zero, so rank seven is maximal here. The subdeterminant as a function of the edge conductance is a nonzero polynomial, giving rank seven for all but finitely many conductance values, although no claim is made for every perturbed weight.

To obtain the actual optimum, in backward time choose p on $[0, a]$, t_2 on $[a, b]$, and t_1 on $[b, \tau]$, where $(a, b, \lambda_1, \lambda_2)$ are the unique certified local solution to

$$\begin{aligned} h_1^\top F(a,b) = h_2^\top F(a,b) = 0, \\ (g + \lambda_1 h_1 + \lambda_2 h_2)^\top P_a (e_p - e_{t_2}) = 0, \\ (g + \lambda_1 h_1 + \lambda_2 h_2)^\top P_b (e_{t_2} - e_{t_1}) = 0. \end{aligned} \tag{25}$$

Here

$$F(a,b) = P_\tau Z + 100\big[K(0,a)e_p + K(a,b)e_{t_2} + K(b,\tau)e_{t_1}\big].$$

The approximate root is

$$\begin{aligned} a &= 0.00513051262657\ldots, \qquad b = 0.00756204696120\ldots, \\ \lambda_1 &= 0.98006094114\ldots, \qquad \lambda_2 = 0.98013504755\ldots. \end{aligned}$$

A four-dimensional rational Taylor/Banach certificate establishes the exact root. On the entire time interval, both relevant dual source-gain differences decrease strictly, and every winning port gain is positive. Thus the maximizing sequence is unique almost everywhere and its exact physical support equals (23). The full support bound with $\lambda_i > 0$ and the terminal equalities proves the asserted optimum, as in Theorem 4.1.

The strict uniqueness of the pointwise winning port gives a singleton exposed face. Equivalently Theorem 5.1 yields $S_v = 0$, because there is no active positive-measure tie. The two fixed-region caps must consequently take the single values (24). Intrinsic connected-region alternatives remain: there are five valid regions for each terminal, three containing b, hence nine source-native pairs with both $\chi_b = 1$ on this unique field. The two forms of uniqueness must not be conflated. □

7.1 A controlled comparison of the two physical regimes

Table 2 summarizes the distinction between the two nearby eight-vertex thermal plants. The comparison changes a single rational conductance; all semantic definitions, heater locations, initial temperatures, and power constraints are held fixed.

Property	Symmetric tree	One-percent perturbed tree
x_2t_2 conductance	1	101/100
Terminal-difference Krylov rank	2	7
Optimal physical endpoint fields	continuum	singleton
Optimal Bright score at $Q = 100$	$10.0721777528649\ldots$	$10.0722304026414\ldots$
Caps for fixed true R_1, R_2	exact diagonal segment	one ordered pair
Both terminal membership flags	attainable	attainable
Distinct intrinsic region choices	retained	retained

Table 2: Controlled physical comparison: the network perturbation changes physical optimal-face geometry, but not the definition of the EF–FG terminal records.

The distinction between the physical fiber and the semantic fiber can also be expressed as a two-stage map. Let $\mathcal{U}$ be the set of admissible controls and $\mathcal{R}(F)$ the set of compatible intrinsic connected-region tuples over the shared F. Then the realized records form the relational image

$$\mathcal{J} = \big\{ (F, \mathrm{ResRel}(F, R_1), \mathrm{ResRel}(F, R_2)) : F = F(u),\ u \in \mathcal{U},\ (R_1, R_2) \in \mathcal{R}(F) \big\}. \tag{26}$$

Its fiber over a physical field can contain several R-choices, even when F itself is unique among optimal controls. Its image under a fixed-region cap map can instead shrink from a segment to a point when spectral ties are removed. Both effects are present in Table 2; neither can be read from a collection of independently attained marginal records.

8 Physical meaning and comparison with existing methods

The application in Sections 4–7 has three concrete levels. At the *state* level, heat diffusion maps a constrained power schedule to a final temperature field. At the *semantic* level, EF–FG extracts ownership and connected terminal records from that same field. At the *joint feasibility* level, the support dual determines whether all requested records can coexist and, in the resource-threshold model, the minimum power required. This is more informative than independently displaying separate temperature fields for each request, which would not establish one common physical witness.

The sharp threshold is actionable inside the mathematical plant model: below Q_* no allowable switching or sharing schedule succeeds; above it a physically specified two-switch schedule does. The fiber statements add a distinct operational message: even when a chosen ownership score is optimal, separately reported terminal caps may be incompatible because their pair is constrained to a diagonal segment. Symmetry breaking can remove the continuum of physical optima without eliminating the plurality of valid semantic region choices.

These results should not be confused with a benchmark demonstrating that EF–FG numerically outperforms standard optimal control. The support dual and switching principles are rooted in established optimal-control theory [3, 4]; the rank test is based on familiar graph control and spectral theory [6, 7]; the nonsemialgebraic proof uses classical real algebraic geometry [9]; and the use of symmetry in optimal control has an independent literature [8]. What EF–FG contributes *within this study* is the specified family of structurally meaningful output predicates and the constraint that all tuples and connected-region choices be evaluated on one physical realization. The resulting sharp combined statements, rather than the component tools, are the object of the present manuscript.

The model has important boundaries. Actuators are ideal and share a pointwise power limit; edge conductances and initial conditions are exact; measurements, finite switching delays, and experimental device imperfections are not incorporated. We prove fixed-time exact realization, not equality of complete trajectories. General finite input EF–FG atlas compilation, complexity bounds for arbitrary irrational spectra, and experimental improvements over classical controllers are not claimed.

9 Conclusion

A network heat equation supports exact mathematical realization of selected EF–FG relational predicates. On one rational nonsymmetric tree, three individually and pairwise feasible semantic queries have a sharp common-power obstruction, resolved by a certified optimal two-switch schedule. On another tree, the actual jointly feasible optimal terminal caps obey a necessary-and-sufficient line-segment law despite a nonsemialgebraic physical support face. A one-percent perturbation of a single conductance yields a finite Krylov-rank jump and a unique physical optimum while leaving multiple admissible terminal resolutions. The common mathematical mechanism is the combination of one shared heat field, exact semantic connectedness and cap constraints, and a global physical support certificate.

Statements and Declarations

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Data availability. No external empirical dataset is used. All network structures, conductances, initial states, switching-time equations, thresholds, and numerical test inputs required for the reported results are specified in this article. No supplementary dataset is necessary.
Code availability. No source code or executable files are deposited or distributed with this submission. The exact-arithmetic verification and independent high-precision numerical checks reported here can be independently reconstructed from the graph Laplacians, heat-kernel integrals, finite-dimensional root systems, rational Taylor remainder bounds, and dual inequalities explicitly

given in the article. No unreleased program is an assumption of any theorem or a prerequisite for reproducing the mathematical certificates.

Ethics approval and consent. Not applicable; this study contains no human or animal subjects.

A Certificate construction and independent reproduction

The manuscript's quoted decimals are display approximations of mathematically defined exact real roots or kernel integrals. Verification consists of certified rational interval arithmetic and exact symbolic calculations; numerical validation consists of independent high-precision heat-flow evaluations using a different representation of the same finite thermal systems. The calculations, not separate executable files, are specified by the following reproducible steps.

A. Semantics and full-control chamber certification. For a Laplacian with $\|L\|_\infty < 6$ and $\tau = 0.01$, evaluate P_s by $\sum_{k=0}^{N}(-sL)^k/k!$ with rational interval arithmetic. Because P_s is nonnegative and stochastic, $\|P_s\|_\infty = 1$ and the truncation errors can be bounded uniformly. Upper and lower interval bounds for all vertex differences, births, maximin attachment levels, and terminal-side component constraints certify the semantic chambers. A separate exact finite enumeration of all connected candidate R_i checks the forced peak, forced terminal, and tie-zone conditions; no disjoint global partition is inserted. For Section 4, the chamber is valid for all pointwise budgets through 109. For Sections 6–7, it is verified for $Q = 100$.

B. Seven-vertex resource threshold. With $q = Q/100$, $A = a/\tau$, $B = b/\tau$, use the rational center $(q, A, B) = (1.083286511336406946, 0.555636232738007475, 0.779845184741640996)$ and a cube of ℓ_∞ radius 10^{-12}. The kernel Taylor truncation through degree 42 has uniform remainder less than $4.79 \cdot 10^{-106}$. A rational approximate inverse B_0 of the Jacobian gives

$$\|\Phi(x_0)\|_\infty < 2.73 \cdot 10^{-19}, \quad \|I - B_0 D\Phi(x_0)\|_\infty < 1.27 \cdot 10^{-12}, \quad \|B_0\|_\infty < 4.$$

With an explicit Hessian bound the Newton-type map $x \mapsto x - B_0\Phi(x)$ has Lipschitz constant below $9.22 \cdot 10^{-11}$ on that cube and maps it into itself. Banach's theorem proves existence and local uniqueness of the exact root. The positive dual multiplier intervals are $(0.989, 0.991)$ and $(0.973, 0.976)$; the two source-gain differences have full-interval strictly negative derivatives, and all winning source gains exceed 0.893. These inequalities were checked using rational enclosures of the matrix-exponential series; the displayed floating-point values are not used to certify their signs. As an independent numerical validation, the original seven-vertex heat equation was also evaluated at 105-digit precision with a separate matrix-Taylor construction.

Constant-control witnesses in Table 1. For a fixed allocation $u = (u_p, u_{t_1}, u_{t_2})$ at every time, set $\widetilde{u} = u_p e_p + u_{t_1} e_{t_1} + u_{t_2} e_{t_2}$. Then the endpoint has an entirely rational Taylor approximation

$$F^{(12)} = \sum_{k=0}^{12} \frac{(-\tau L)^k}{k!} Z + \sum_{k=0}^{12} \frac{(-L)^k \tau^{k+1}}{(k+1)!} \widetilde{u}.$$

For the seven-vertex graph $\|L\|_\infty < 6$, $\tau = 0.01$, $\|Z\|_\infty = 25$ and $\|\widetilde{u}\|_\infty \le 100$. The matrix-exponential remainder and its time integral give

$$\|F - F^{(12)}\|_\infty \le (25 + 100\tau) e^{6\tau} \frac{(6\tau)^{13}}{13!},$$

so each displayed two-vertex contrast differs from the corresponding rational Taylor value by less than $1.2 \cdot 10^{-24}$. The four rows in Table 1 are the correctly rounded five-decimal evaluations of these rational expressions. Their strict sign margins are therefore certifiable without relying on floating-point signs.

C. Symmetric eight-vertex optimum and cap endpoints. Degree-33 exact rational Taylor bounds have remainder below $9.73 \cdot 10^{-81}$. They bracket $0.00512997793228 < s_* < 0.00512997793230$, certify $1.959 < \lambda_* < 1.962$, and establish strict source-gain monotonicity throughout $[0, 0.01]$. Exact symbolic calculation on the arm-swap antisymmetric space produces eigenvalues $(3 \pm \sqrt{5})/2$. An independent 116-digit spectral-integration evaluation also reproduced the root, the switching median c, and the endpoint cap difference δ; this numerical check is distinct from the analytic proof of nonsemialgebraicity.

D. Perturbed eight-vertex optimum. A direct rational Laplacian Krylov calculation verifies the displayed 7×7 minor (rows 2–8 counted from one, not rows 1–7). The four unknowns $(a/\tau, b/\tau, \lambda_1, \lambda_2)$ have the rational center

$$(0.513051262657132927,\ 0.756204696120191786,\\ 0.980060941142877874,\ 0.980135047554554104)$$

and a 10^{-12} local cube. The degree-34 matrix-Taylor error is below $1.67 \cdot 10^{-83}$, the rational contraction constant below $2.08 \cdot 10^{-10}$, and all source-gain ordering inequalities hold over the *entire* time interval. An independent 115-digit direct spectral evaluation reconstructed the root and cross-checked the EF–FG connected-region classification and the physical field.

E. Reproduction of the verified quantities without external code. An independent calculation begins with the rational edge list and the initial vector specified for each graph in Sections 4 and 6–7. Form the weighted Laplacian $L_{ii} = \sum_{j \neq i} c_{ij}$, $L_{ij} = -c_{ij}$, and approximate $P_s = e^{-sL}$ using

$$P_s^{(N)} = \sum_{k=0}^{N} \frac{(-sL)^k}{k!}, \qquad \|P_s - P_s^{(N)}\|_\infty \le e^{s\|L\|_\infty} \frac{(s\|L\|_\infty)^{N+1}}{(N+1)!}.$$

Integrate this polynomial term by term over each stated switching interval, retaining rational lower and upper enclosures for all entries. For the seven-vertex system, solve its three equations for the two switches and Q, check the contraction estimates in part B, and then verify the pointwise dual-port ordering on $[0, \tau]$. For the symmetric eight-vertex model, solve the terminal equality and the scalar switching median equation, evaluate both cap endpoints, and check the interval bounds of part C. In the perturbed model, solve the four equations for two switches and two dual multipliers, verify the rational Krylov minor, and check the uniform source-order inequalities in part D.

The remaining EF–FG verification is finite: enumerate connected subsets containing each terminal, reject subsets violating the peak-side exclusion or forced-terminal inclusion, and calculate their floors, caps, query values, and membership flags from the *same* terminal field. The numerical validation reported above uses independent high-precision evaluation of the original heat equation to cross-check the certified quantities, without replacing the exact inequalities by numerical tests. These specifications are sufficient to write independent implementations; no software artifact or external dataset is required to interpret or rederive the claims. Hardware or laboratory validation is not claimed.

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